Vidhyadhan Charitable Trust

# SMT Z.S Patel College of Computer Application

Affiliated to, Veer Narmad South Gujarat University, Surat

# Social Media Usage & Screen Time Analysis

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# Introduction

- Social media is widely used for communication and entertainment.
- People spend more time on screens daily.
- Excessive screen time affects:
    1. Sleep quality
    2. Mental health
    3. Productivity
- This project analyzes user data to understa these effects.

# Project Goals & Objectives

- Analyze dataset of **50,000 users.**
- Identify usage patterns across age and occupation.
- Study relation between:
    1. Screen time
    2. Stress levels
    3. Sleep duration
- Build ML models to:
    1. Predict phone usage
    2. Detect high stress users

# Dataset – Overview

- Dataset file: **socialmedia_data.csv**
- Total Records: **50,000**

**Includes:**

- **Demographics:** Age, Gender, Occupation
- **Usage:** Daily phone hours, social media usage
- **Lifestyle:** Sleep hours, stress level
- **Device Type:** Android, iOS, PC

| User_ID | Age | Gender | Occupation | Device_Type | Daily_Phone_Hours | Social_Media_Hours |
|---|---|---|---|---|---|---|
| U1 | 58 | Male | Professional | Android | 1.3 | 6.7 |
| U2 | 25 | Male | Professional | Android | 1.2 | 1.5 |
| U3 | 19 | Male | Student | iOS | 5.3 | 5.7 |
| U4 | 35 | Female | Business Owner | iOS | 5.8 | 2.5 |
| U5 | 33 | Male | Freelancer | Android | 7.9 | 1.3 |
| U6 | 32 | Female | Student | Android | 10.9 | 4.2 |
| U7 | 26 | Male | Student | iOS | 5.6 | 6.1 |
| U8 | 24 | Male | Freelancer | iOS | 8.5 | 1.9 |
| U9 | 52 | Female | Business Owner | Android | 9.4 | 2.5 |
| U10 | 23 | Female | Student | iOS | 2.8 | 1.1 |
| U11 | 55 | Other | Professional | Android | 8.3 | 1.1 |
| U12 | 45 | Other | Business Owner | Android | 12 | 6.7 |
| U13 | 20 | Female | Business Owner | iOS | 11.1 | 5.6 |
| U14 | 19 | Female | Freelancer | iOS | 9.7 | 4.1 |
| U15 | 23 | Male | Professional | Android | 1.1 | 2.5 |
| U16 | 31 | Male | Business Owner | Android | 11.6 | 5.5 |
| U17 | 32 | Other | Professional | Android | 2.3 | 7.8 |
| U18 | 50 | Female | Student | Android | 10.8 | 5.7 |
| U19 | 56 | Female | Business Owner | Android | 3.3 | 4.3 |
| U20 | 19 | Other | Professional | Android | 3.9 | 4.2 |
| U21 | 53 | Other | Business Owner | Android | 3.3 | 2.6 |
| U22 | 30 | Other | Professional | iOS | 9.5 | 6.9 |
| U23 | 59 | Female | Freelancer | Android | 10.9 | 4.9 |
| U24 | 52 | Female | Student | iOS | 5.5 | 5.1 |
| U25 | 44 | Female | Business Owner | Android | 3.7 | 4.2 |
| U26 | 32 | Female | Professional | iOS | 10.1 | 7.5 |
| U27 | 46 | Male | Business Owner | iOS | 1.2 | 6.5 |
| U28 | 55 | Other | Student | iOS | 9.7 | 6.1 |
| U29 | 35 | Male | Professional | iOS | 11.7 | 1.6 |
| U30 | 18 | Other | Student | iOS | 11.4 | 7.8 |
| U31 | 28 | Other | Student | Android | 2.1 | 2.3 |
| U32 | 45 | Other | Freelancer | Android | 5.8 | 3.8 |
| U33 | 39 | Other | Professional | Android | 8.2 | 2.3 |
| U34 | 35 | Other | Freelancer | iOS | 3.4 | 2 |
| U35 | 27 | Female | Student | Android | 11.8 | 6.3 |

# Methodology & Tools

**Python**

**Libraries Used:**

- Pandas → Data cleaning & analysis.
- Matplotlib & Seaborn → Data visualization.
- Scikit-learn → Machine Learning.

**Tools:**

- VS Code
- Jupyter Notebook
- Google Colab

# Data Cleaning & Preprocessing

- Removed missing values (**NULL**).
- Deleted duplicate records using **User_ID**.
- Converted data into proper format.
- Handled outliers for accurate results.

# Feature Engineering

**What is Feature Engineering?**

- Feature Engineering is the process of **creating new features (columns)** from existing data.
- It helps improve **data analysis and machine learning performance**.
- Converts raw data into **meaningful insights**.

**Benefits of Feature Engineering:**

- Improves **model accuracy.**
- Makes data easier to understand.
- Helps find **hidden patterns.**
- Enables better **predictions.**

# Estimated Weekly Screen Time

- A new feature to calculate **total weekly usage.**

**Why this is useful?**

- Gives a **complete view** of user screen time
- More accurate than just daily usage

**Example:**

- Daily = 4 hours
- Weekend = 6 hours
- Weekly = (4×5) + (6×2) = **32 hours**

**Formula Used:**

Weekly Screen Time = (Daily Hours × 5) + (Weekend Hours × 2)

# High Stress Indicator (Binary Feature)

- Created a new column to identify **high stress. users.**

**Condition:**

- Stress Level ≥ 7 ⟶ **High Stress (Yes).**
- Stress Level < 7 ⟶ **Normal (No).**

**Why this is useful?**

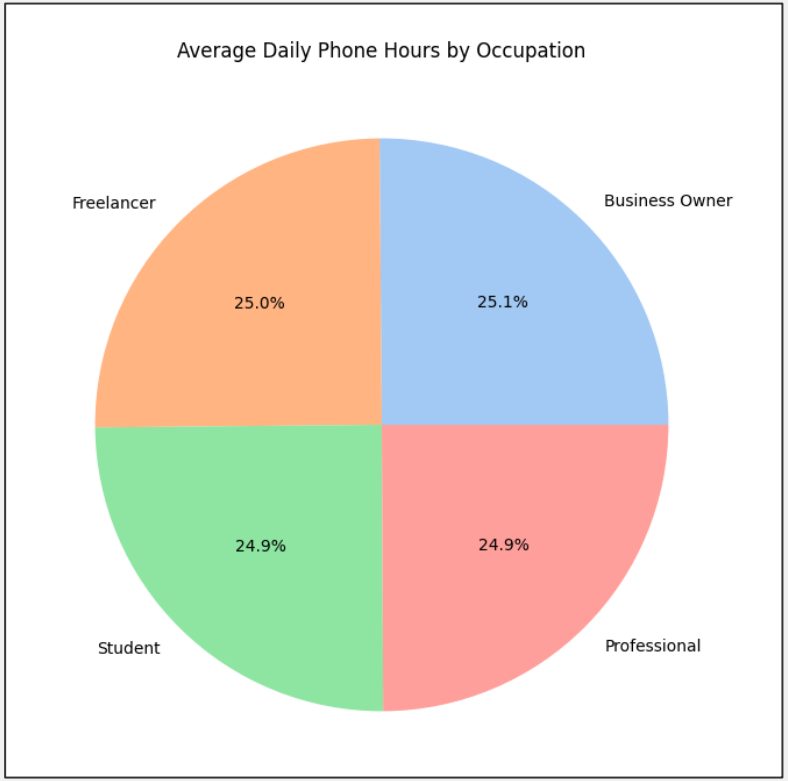
- Converts complex data into **simple categories**
- Helps in **classification model (ML)**

**Example:**

- Stress = 8 ⟶ High Stress.
- Stress = 5 ⟶ Normal

# EDA 1 – Occupation vs Phone Usage

- Compared phone usage across occupations.
- Students and IT professionals use phones the most



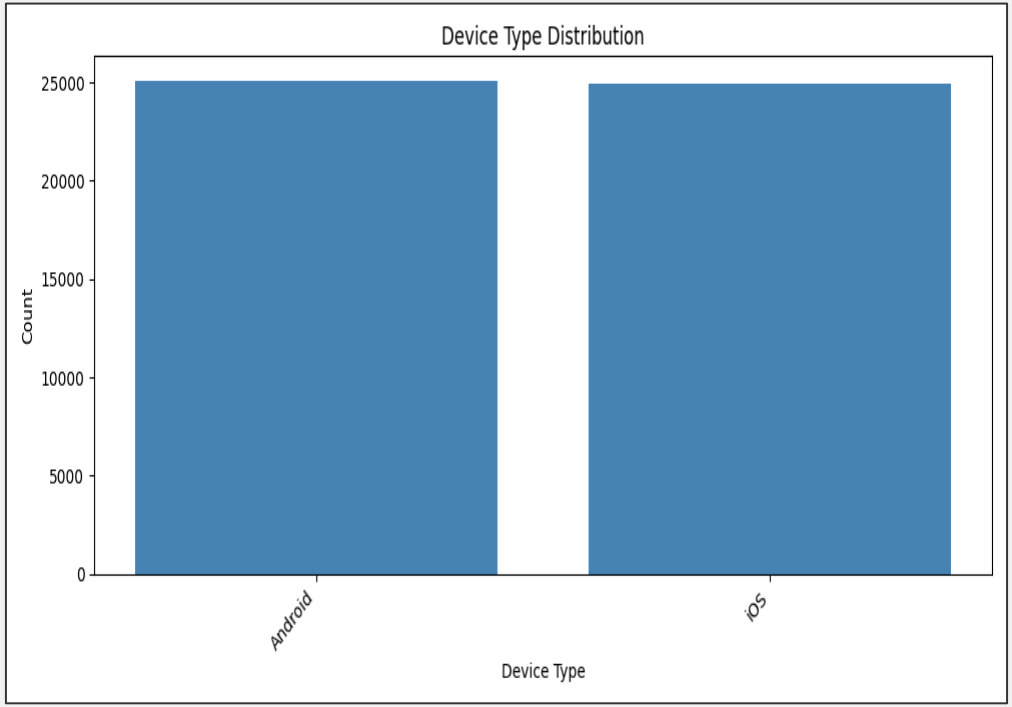
# EDA 2 – Device & Age Trends

- **Device usage comparison:**

1. Android vs iOS vs PC.

- **Age-based trends:**

1. Young adults have highest usage.

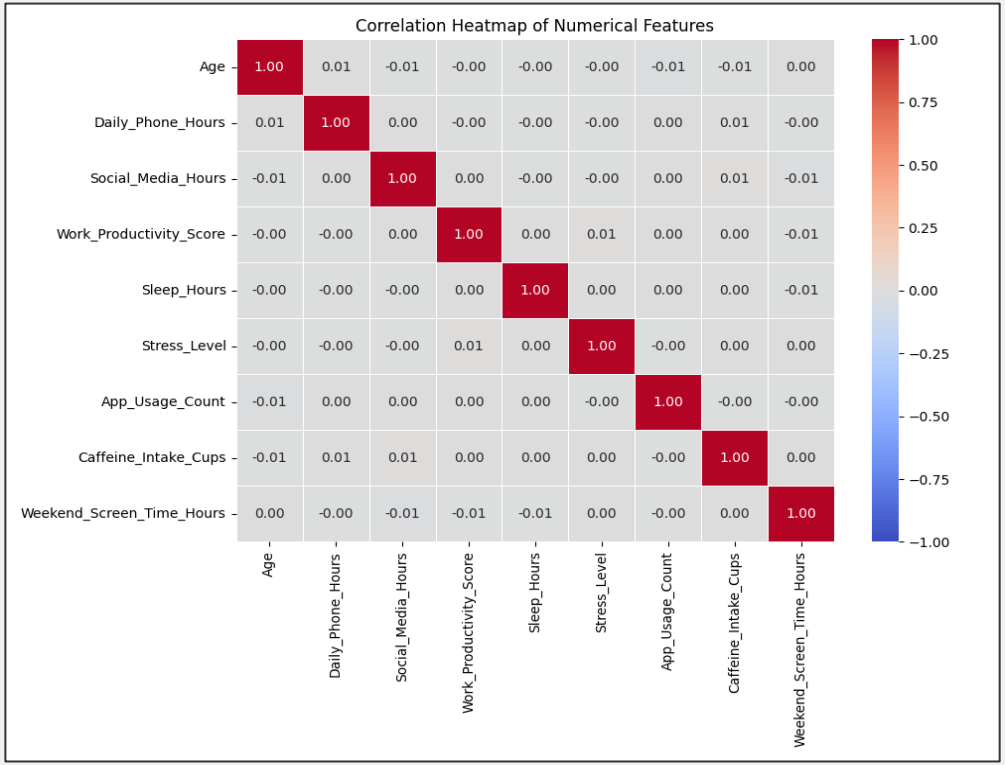


# EDA 3 – Correlation Heatmap

- Shows relationship between variables.

**Key Insights:**

- More screen time ⟶ Less sleep.
- More social media ⟶ Lower productivity.
- Higher usage ⟶ Higher stress.



# Machine Learning – Regression

**Goal:** Predict daily phone usage.

**Model:** Linear Regression.

# Evaluation Metrics

Mean Squared Error (MSE).

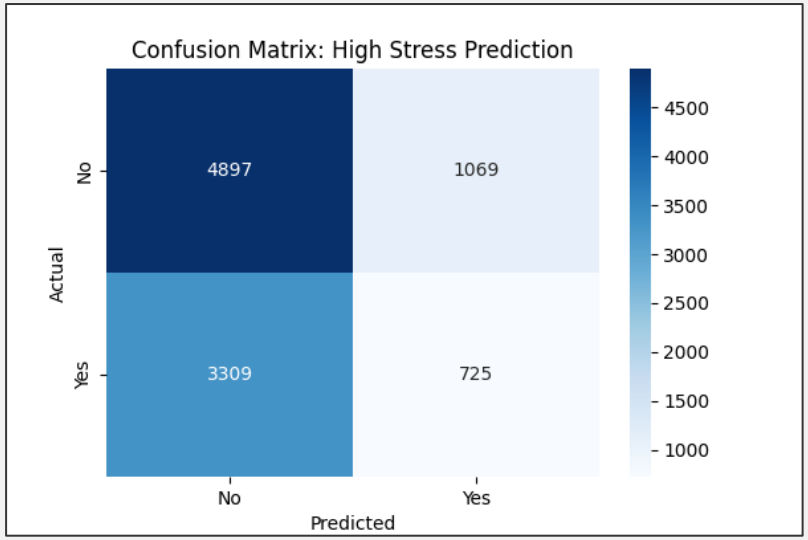
$R^2$ Score.

# Machine Learning – Classification

**Goal:** Identify high stress users.

**Model:** Random Forest.

**Output:** Yes / No (High Stress)



# Major Findings

High social media usage → Less sleep.

Weekend screen time is higher.

Heavy users show more stress.

# Future Scope

Real-time screen tracking using apps.

**Smart Alerts for**:

- Reducing screen time.
- Improving mental health.

Integration with health monitoring systems.

# Conclusion

- Screen time affects lifestyle and health.
- Data analysis helps understand user behavior.
- Can help people build healthier digital habits.

# Thank You!